

Simone Carletti

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About Me

I am a **Computer Engineering M.Sc.** student specializing in **Robotics and Autonomous Systems**. I'm currently completing my master's thesis at **EPFL**, where I am developing a control system for a **morphing-wing drone** using **Reinforcement Learning** and **$\mathcal{L}_1$ Adaptive Control**. I have hands-on experience across the robotics stack, from **perception** and **planning** to **control systems** and **embedded hardware**, with proficiency in **C++**, **Python**, **ROS2**, and **TypeScript**. I am experienced in transitioning algorithms from Python/Simulink simulations to high-performance embedded implementations (C/Rust). Beyond my research at **EPFL** and **DTU**, my **professional experience** as a **lead developer** has taught me to write clean, maintainable, and production-ready code.

Education

EPFL - LIS , <i>Master's Thesis</i>	<i>Sept 2025 – Present</i>
- Title: L1-Adaptive augmentation for robust morphing-wing drone flight control - Skills: Control Systems, RL (PPO), C++, Python, Simulink, ROS, Git, Linux, Unity	
DTU , <i>M.Sc. Autonomous Systems</i>	<i>Sept 2024 – Jul 2025</i>
Erasmus+ exchange - Converted GPA: 30/30	
Politecnico di Torino , <i>M.Sc. Computer Engineering</i>	<i>Sept 2023 – Present</i>
Curriculum in Automation and Intelligent Cyber-Physical Systems - Final GPA: 29.48/30	
Università degli Studi di Firenze , <i>B.Sc. Computer Science and Engineering</i>	<i>Sept 2020 – Sept 2023</i>
- Graduated with 110/110 cum laude (with honors) - Thesis: Sensor network for monitoring the greenhouses of the Botanical Garden of Florence	

Experience

Robotics Engineer , <i>DIANA student team</i>	<i>Sept 2023 – Jul 2024</i> <i>Turin, Italy</i>
Collaborated on a Mars rover student project, implementing perception and EKF state estimation in C++ while contributing to software development, testing, and system integration	
Full-Stack Developer , <i>Genomedics S.r.l</i>	<i>Sept 2020 – Present</i> <i>Florence, Italy</i>
- Lead developer of three key projects that handle the release, deployment, and licence management for all the company's products. Developer for the company's main project (KPlatform) - Skills: Angular (Typescript, SCSS, HTML), Vue.js, PHP, Laravel, SQL, Python, Git, Linux	

Selected Projects

Autonomous Navigation & Task Execution - DTU RoboCup 🔗	<i>Feb 2025 – May 2025</i>
Developed autonomous navigation, control, and task execution algorithms with continuous integration and validation tests on the real platform	
Digital Control - Autonomous Robotics (GitHub 🔗)	<i>Oct 2024 – Dec 2024</i>
Designed digital controllers and implemented them on embedded hardware using C++	
3D Object Tracking using Stereo Camera	<i>Nov 2024 – Dec 2024</i>
Developed a 3D perception pipeline integrating Faster R-CNN for detection, SIFT-based stereo imaging, and EKF for robust object tracking (GitHub 🔗)	
Comparison and implementation (from scratch) of YOLOv1 and U-Net	<i>Oct 2024 – Dec 2024</i>
Implemented landmark computer vision models from scratch using PyTorch (GitHub 🔗)	
Multiplayer videogame on embedded hardware	<i>Jan 2024 – Jun 2024</i>
Developed bare-metal C firmware on ARM Cortex-M3 with a custom CAN protocol (GitHub 🔗)	
Localization with EKF in ROS2	<i>Nov 2023 – Dec 2023</i>
Implemented a full EKF localization pipeline from scratch in ROS2 and Python, fusing data from LiDAR, IMU, and wheel odometry	

Skills & Activities

- Languages:** Fluent in English, Native Italian speaker, Basic Spanish skills
- Programming:** C++, C, Python, TypeScript, JavaScript, Rust, Java, Matlab, PHP, SQL, HTML, CSS/SCSS, Assembly, LaTeX
- Tools:** PyTorch, Linux, Git, Google Colab, OpenCV, Onshape, CMake, Docker, Node-RED, Multisim
- Other:** Raspberry Pi, Arduino, ESP8266, NodeMCU, Soldering, Nvidia Jetson/Orin, Guitar, Photography, Networking, Zigbee
- Volunteering:** ESN Copenhagen and DTU (Organized events, supported international students and promoted cultural exchange)